

Digital plaque monitoring: An evaluation of different intraoral scanners

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ABSTRACT

Objectives: Intraoral scanners (IOS) display disclosed plaque, and the scientific literature has reported that plaque levels can be monitored on intraoral scans using one IOS system (Dexis 3800; control IOS). This study aimed to investigate whether this is also possible with other IOS systems (i700, Primescan, Trios 5; test IOS).

Materials and methods: Ten participants (29.6 ± 5.5 years) were enrolled. After plaque accumulation and subsequent toothbrushing, intraoral scans were performed with the control IOS and the three test IOS. All scans were aligned and the vestibular/oral surfaces of the Ramfjord teeth (16, 21, 24, 36, 41, 44) were analysed with automated planimetry using a predefined threshold value. The proportion of pixels assigned to plaque-covered areas was expressed as a percentage of the total number of pixels (P%). We then assessed whether the planimetrically determined plaque-covered areas corresponded to those identified visually. This revealed that a threshold correction (P%_{corr}) was required for approximately 20 % (i700 and Trios 5) to over 65 % (Primescan) of the images.

Results: Bland-Altman analysis showed no significant systematic bias and limits of agreement ranging from approximately -20 to +20 P% units, with a tendency towards lower values at higher plaque coverage. Manual correction improved the agreement and halved the limits of agreement. All test IOS could detect a reduction in plaque after brushing, as well as the typical site-dependant plaque distribution patterns.

Conclusions: All test IOS appeared to be suitable for plaque monitoring. Planimetric methods must be adapted to the colour representation of the IOS.

Clinical significance: Plaque monitoring using IOS opens a new field of application in preventive dentistry.

1. Introduction

Plaque detection, usually after disclosing with a staining agent, has a long tradition in dentistry [1]. To date, various index systems have been used, in which the tooth surface is divided into different areas that are then scored for the presence or absence of plaque. However, this clinical approach has the disadvantages that (i) plaque conditions are not captured on images, (ii) some index systems are difficult to apply, especially with regard to the different anatomical shapes of different tooth types and tooth surfaces, (iii) examiner calibration is demanding, and (iv) the outcome of interventions [2] may be influenced by the plaque index used. In addition, not all plaque indices are equally effective in representing true plaque coverage [3] and plaque growth [4].

An alternative method that does not have these disadvantages is the

planimetric measurement of plaque, which is an objective method for plaque quantification. In patients with inflamed gingiva, planimetry appears to have greater discriminatory power than clinical indices [4]. Publications on this method date back to the early 1980s, when the proportion of the total tooth surface covered by plaque was determined manually, often using planimeters [5] or densitometers [6]. This method was further developed using computer-assisted imaging [7–11]. The evaluation was mostly based on images taken with extraoral cameras, which have a major disadvantage in that the tooth surfaces, especially the lateral teeth, are often not fully displayed or are not homogeneously illuminated because of an unfavourable imaging angle. Consequently, shadows are mostly visible on the mesial and distal parts of the tooth, making it difficult to distinguish between pixels representing a clean surface and those representing a plaque-covered surface. In addition, it

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is technically difficult to generate high-quality images of the lateral teeth using extraoral cameras. Therefore, planimetric methods have often been limited to the anterior teeth [12,13]. The use of intraoral cameras has partially solved these problems [9,11]; however, capturing all tooth surfaces of the dentition is remains time consuming.

Intraoral scanners (IOS) with colour imaging alternatively provides uniformly illuminated three-dimensional (3D) images of the entire dental arch in a relatively short time, thus making it easier to evaluate and determine plaque conditions. Another advantage is that 3D scans can be digitally aligned with multiple follow-up scans and then rotated in space, thus eliminating the problem of standardising the camera orientation relative to the tooth surface. In addition, each tooth projection can be optimally adjusted for all aligned scans as required.

IOS have been applied as a method for oral hygiene instruction using a plaque index [14] or for evaluating the cleaning effects of interdental brushes using a planimetric method [15].

Our previous study showed that the Rustogi modified Navy-Plaque-Index obtained on IOS images provides comparable results to those obtained from intraoral camera images and clinical examination [11]. Another study showed that planimetric plaque evaluation from 2D images generated from 3D scanner images corresponded well with planimetric measurements from 2D intraoral camera images [16].

There are many types of IOS systems that are increasingly being used in practice and research for diagnostic purposes, such as caries diagnostics and wear measurement [17]. In addition to the standard application of digital impressions, it would be of interest to add plaque quantification to these functions. However, IOS systems differ in terms of acquisition technology [17], which affects image quality and colour representation; therefore, the results of our previous study [16] may not be easily extrapolated to other IOS types.

This study aimed to determine the agreement between the planimetrically assessed amounts of plaque recorded by three types of IOS (test IOS) using different acquisition and illumination technologies and a control IOS that has been used in a previous study [16]. The null hypothesis was that the plaque quantified using the test IOS would not differ from those quantified using the control IOS.

2. Participants, materials and methods

2.1. Study design

This methodological study was approved by the local Ethics

Committee of the Medical Faculty (ref. no. 61/22) and was conducted in accordance with good clinical practice and the Declaration of Helsinki. The study design is illustrated in Fig. 1.

2.2. Patient recruitment

Participants were recruited from our staff and dental students. Inclusion criteria were age 18 years or older, written informed consent provided, and complete dentition (except third molars) without restorations on vestibular and oral smooth surfaces beyond the extent of pit and fissure sealants. Exclusion criteria were intolerance to the materials used (Mira-2-Ton® solution: sodium benzoate, potassium sorbate, C. 45,410, C. 42,090; OptraGate lip retractor: styrene-ethylene-butylene-styrene), cavitated caries, fillings with defects on the surface to be examined that facilitate plaque retention, malocclusion, gap teeth and gingival recession of more than 1/3 of the root length, fixed orthodontic appliances (retainers allowed), and physical or mental limitations of any kind that could affect performance.

2.3. Procedure

Ten participants (7 women, 3 men, age 29.6 ± 5.5 years) were included. The participants were asked not to brush their teeth for 72 h before the first appointment and to bring their brushing utensils from home to the next appointment. At the first examination time point after 72 h without habitual oral hygiene (T1), plaque was disclosed with a plaque revelator (Mira-2-Ton®, Hager & Werken GmbH & Co KG, Duisburg, Germany), which was applied twice with a saturated foam pellet. Subsequently, the participants rinsed with water for 10 s after each application. After disclosing, an intraoral scan was performed using one of the four IOS systems (Table 1) in randomised order. A randomisation list was generated using Research Randomizer version 4.0 [18] prior to the start of the study.

The scanning procedure was performed under standardised conditions: Cheeks and lips were fixed with a retractor (OptraGate; Ivoclar Vivadent, Schaan, Liechtenstein), and the dental arch was kept dry with parotid patches (DryTips®, Microbrush, Algonquin, IL, USA), a saliva ejector, and an air blower. To minimise washout of the staining agent, the lower jaw was scanned first, followed by the upper jaw. All scans were performed in daylight and all artificial lighting was turned off. Before the next scan with a different IOS, the plaque was stained twice to achieve good colour saturation.

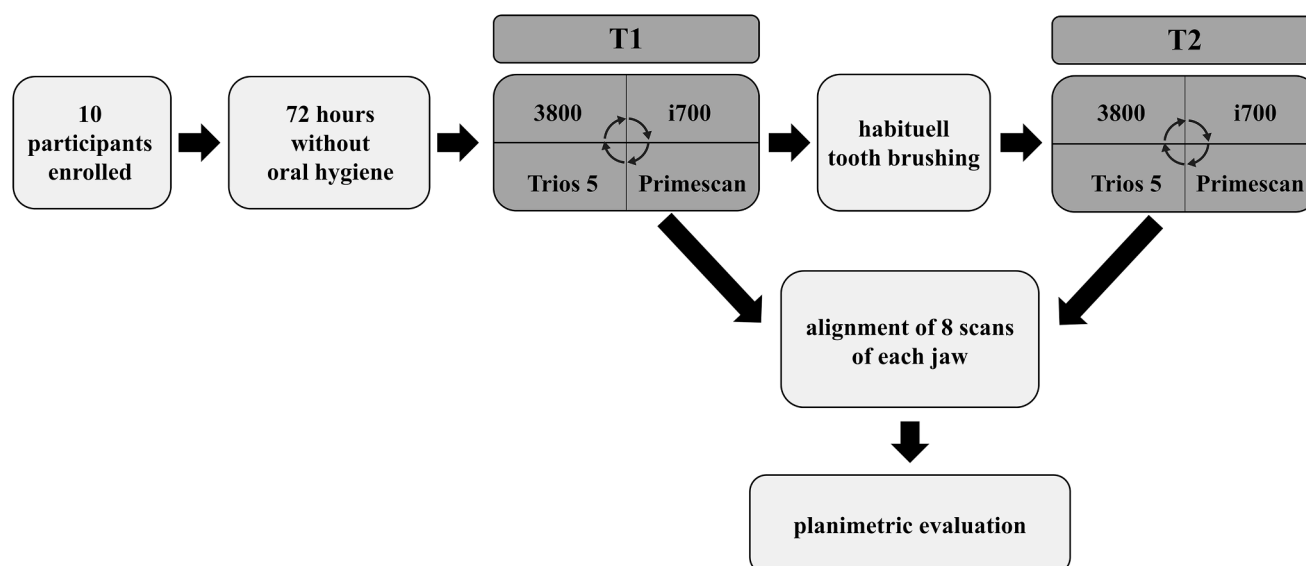


Fig. 1. Flow chart of the study design.

Table 1
Specification, Abbreviation used in the text, and manufacturer’s information for the IOS.

Group	Intraoral scanner	Abbreviation	Software Version	Manufacturers	Technology
control IOS	Dexis IS 3800 W	3800	Dexis 1.0.7.932.d104	Dexis, Quakertown, USA (analogue to Carestream 3800, Carestream LLC, Atlanta, USA)	optical triangulation
test IOS	Medit i700	i700	Medit (2.6.2) i700 Scan (1.8.2)	Medit, Seoul, South Korea	optical triangulation
	Primescan	Primescan	Cerec SW 5.2.4.276189	Dentsply Sirona, Bensheim, Germany	high-frequency contrast analysis
	Trios 5	Trios 5	1.18.5.52 (22.1.6.0)	3Shape, Copenhagen, Denmark	confocal microscopy

After scanning with all four IOS systems, the participants were asked to brush their teeth using their personal toothbrushing equipment. Immediately afterwards, the same procedure as that for T1 was performed at examination time point two after habitual oral hygiene (T2). The order of the IOS was maintained for each participant.

2.4. Plaque quantification through planimetry and outcome variables

The intraoral scans were saved as polygon files (PLY) and the Primescan as OBJ files. The files were processed using MeshLab [19]. All scans of each participant’s jaw (eight scans in total, four different IOS systems at two different times; see Fig. 2) were aligned using at least 14 different points per scan. After alignment, the scans were simultaneously rotated around the x, y, and z axes to provide an optimal view of the tooth surface under examination, ensuring that both mesial and distal parts of the tooth surface were equally visible. Finally, automated screenshots (PNG files) were generated from all scans with a black background. The alignment of the scans and automatic creation of screenshots ensured that the tooth surfaces to be evaluated were identical for all the participant scans at both time points.

The PNG images were then imported into an image processing program (Photoshop CS5 extended, version 12.0 × 64, San Jose, California, USA), and the tooth surface of interest was cropped, placed on a black background, and saved as a JPG image. The tooth section template was applied to all scans at both time points such that the section of the tooth surface to be evaluated did not vary between different IOS or between the two time points. For plaque quantification, we used the software

described in detail in Jung et al. [16] using the Julia programming language and the “ImageMagick” and “Images” packages. The software converted the JPG image into an RGB image. The red, green, and blue values of each pixel were determined, transformed into a 3-component vector system, and stored separately. Two filters were used to calculate the plaque coverage. The first filter removed all black pixels from the black background (RGB = 0,0,0), whereas the second filter allowed a threshold value to be set to distinguish pixels belonging to the plaque-covered tooth surface from those belonging to the plaque-free tooth surface. As the exposure of 3D intraoral scans is constant, the settings could be used in a standardised manner, which allows batch processing.

The percentage of plaque-covered areas (P%) on the total tooth surface was calculated as the ratio of all pixels to the pixels below the threshold. The thresholds for the three test IOS systems were set by visually optimising the plaque coverage on the original image. This was performed for each IOS system, and the researchers were blinded to the results of the other IOS systems. Through batch processing, the values (P %) were automatically generated as an Excel file. Subsequently, the areas automatically identified as plaque-covered were visually compared with the input images. In cases of obvious mismatches, the threshold value for the corresponding image was individually adjusted to achieve the best possible match (P%corr). This was performed prior to further statistical analysis, and the researchers were blinded to the images from the other IOS. This adjustment produced another datasheet with corrected values.

Fig. 3 shows an example of the colour representation of the different

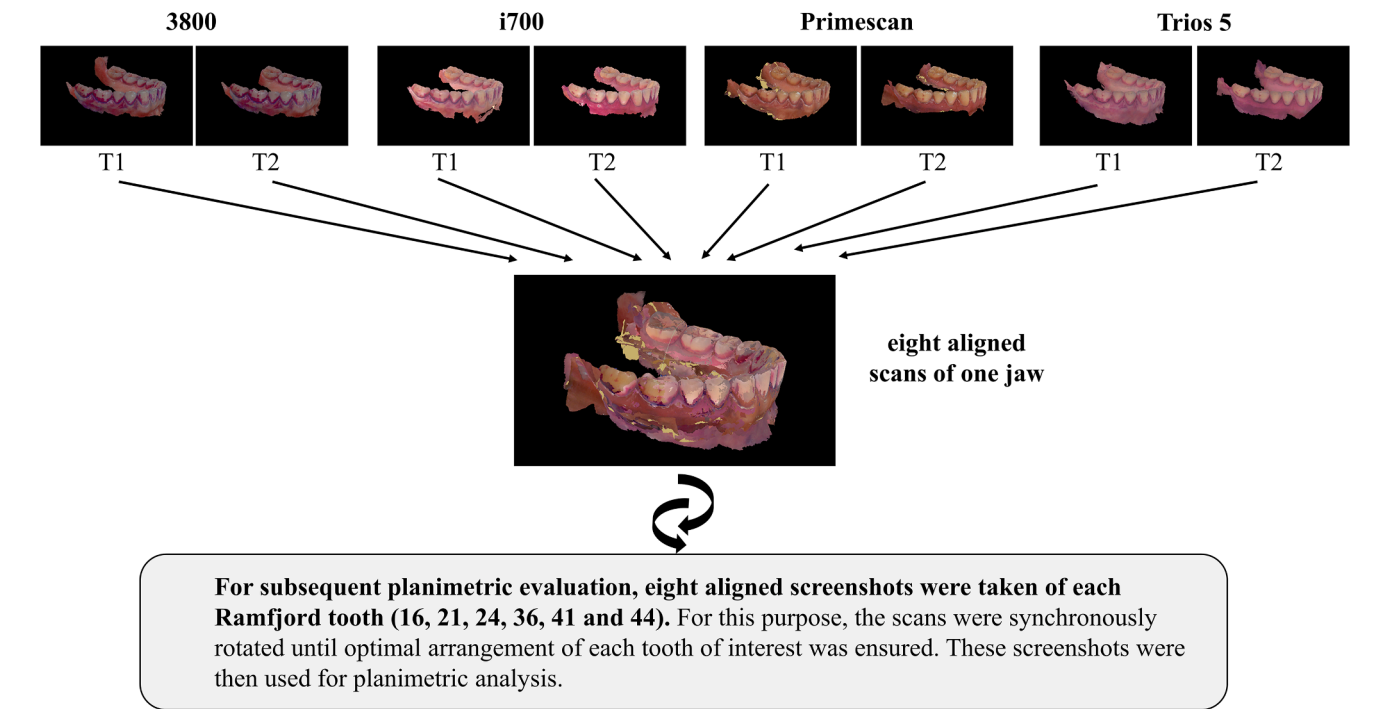


Fig. 2. Graphical representation of the origin of the eight aligned scans and further procedures of the study.

IOS systems as well as a cropped tooth with the corresponding planimetric analysis.

2.5. Statistical evaluation

The sample size calculation was based on preliminary tests with three individuals (data not included in the present study) and on data from a previous study [16], and was performed using MedCalc version 20.251 (MedCalc Software, Ostend, Belgium).

Statistical analysis was performed on two levels:

Tooth level: The unit of observation were the six Ramfjord teeth [20], resulting in 12 observations (the oral and vestibular surfaces of these teeth) per participant.

Participant level: The observation unit was the participant, and the mean value was calculated from the 12 values of the Ramfjord teeth, resulting in one observation per participant.

As there was no agreed value for a clinically relevant difference in the agreement of the measured P% values, a permitted difference of 10 P% units was defined as clinically relevant for the sample size calculation (Fig. 4).

Analysis at the tooth level: The preliminary tests showed a mean (SD) of the differences of the test IOS to the control IOS of 0.45 ± 3.56 for the P% values. This resulted in a minimum sample size of 52 for Bland-Altman analysis, with an α of 0.05 and β of 0.02, and a permitted difference of 10 P% units. With 12 comparisons per participant, this would require a minimum of five participants.

Analysis at the participant level: The preliminary tests showed a mean value (SD) of the differences between the test IOS and the control IOS of 0.45 ± 2.02 for the P% values. This resulted in a minimum number of eight participants for Bland-Altman analysis with an α of 0.05 and β of 0.02, and a permitted difference of 10 P% units.

The study also evaluated the ability of the test IOS to detect a reduction in plaque coverage after 72 h of plaque growth through toothbrushing. Data from the previous study [16] showed a plaque reduction (mean $\pm$ SD) of 6.7 ± 4.1 %. This resulted in a minimum sample size of five participants for *t*-tests for paired samples with an α of 0.05 and β of 0.02.

As there may be unknown participant-related factors that could influence scan quality (e.g. oral space and colour conditions or shape of the dental arch), 10 participants were included, which was slightly larger than the calculated sample size.

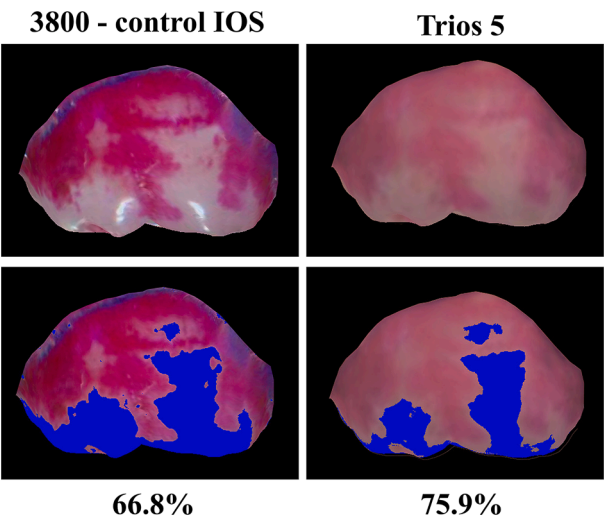


Fig. 4. Illustration of the defined clinically relevant difference for the agreement of the plaque coverage (P%) determined by the test and control IOS. The images show a cropped tooth from the available data (above) and the corresponding planimetric analysis (below). The values for plaque coverage are 66.8 P% for the control IOS and 75.9 % for Trios 5, a difference of approximately 10 P% units.

Statistical analyses were performed using MedCalc version 20.251 (MedCalc Software) and SPSS version 28.0 (IBM, Ehningen, Germany). As the data showed no significant deviation from the Gaussian distribution (Kolmogorov-Smirnov-test), parametric procedures were used. Values are presented as mean $\pm$ standard deviation (SD). Bland-Altman analyses were performed to determine the extent to which the measurements obtained with the test IOS agreed with those of the control IOS. To analyse whether the test IOS could represent plaque reduction after toothbrushing, *t*-tests for paired samples were performed at both the tooth and participant levels. When examining the P% values before and after oral hygiene by tooth group and tooth surface, significant deviations from the Gaussian distribution were observed. Therefore, these values are presented as median with 95 % confidence intervals (after bootstrapping), and the Wilcoxon test for paired samples was used.

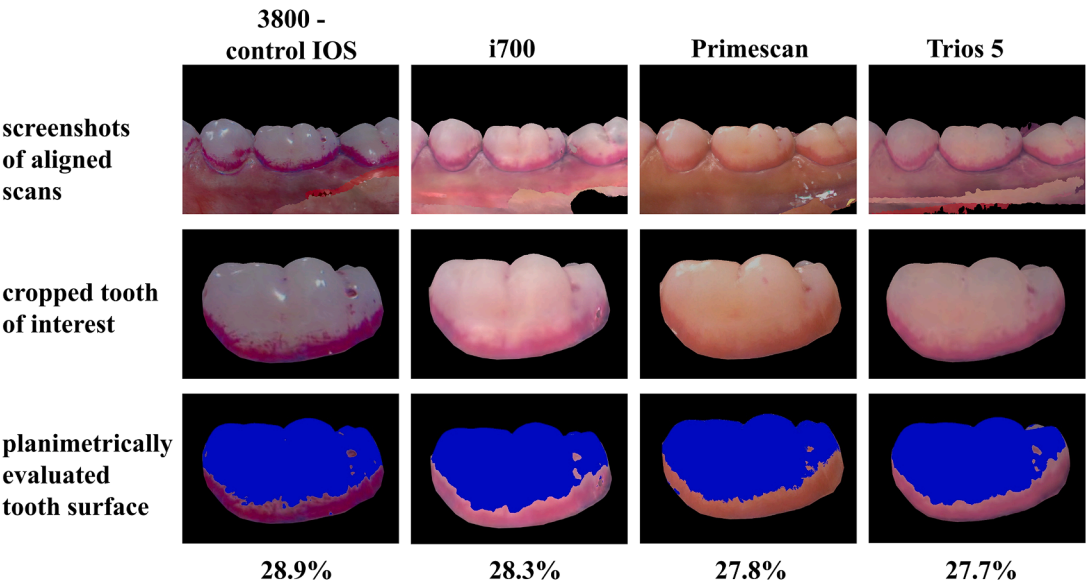


Fig. 3. Colour representation of the different IOS systems and the subsequent planimetric evaluation. Sections of all four IOS are shown, as well as a representation of the cropped tooth of interest and the planimetric evaluation.

3. Results

3.1. Agreement of P% values from the test IOS with the control IOS

At the tooth level, the different IOS systems revealed mean P% values between 30.1 and 32.8 at T1 and 16.3 and 20.7 at T2 (Table 2). Threshold correction was applied to 6.7 % of the teeth at T1 with the control IOS (3800) and to 19.2 %, 70.0 %, and 17.5 % of the teeth with the i700, Primescan, and Trios 5, respectively. The corresponding T2 values are 0.8 %, 10.0 %, 65.8 % and 9.2 %, respectively. After correction, the P%_{corr} values for T1 were between 31.1 and 32.3 and for T2 between 16.8 and 19.2 (Table 2). The Bland-Altman analysis showed no significant bias for the uncorrected values for Primescan and Trios 5; for the i700, there was a minor but significant systematic bias with somewhat lower values. However, for all test IOS, a significant trend was observed for lower values at higher plaque coverage levels compared to the control IOS (Fig. 5, Table 3). The limits of agreement ranged from -21.3 to -19.0 for the lower limit and 19.9 to 21.4 for the upper limit. For Primescan, a different distribution of point clouds for the oral and vestibular readings was visually apparent. Therefore, separate Bland-Altman analyses were performed for this IOS. This showed that the overall trend was due to the vestibular readings alone, which revealed a very strong significant ($p < 0.001$) trend for lower scores at higher plaque coverage levels, whereas the oral readings showed no trend.

After correction, considerably improved agreement with the control IOS was achieved for all test IOS. A significant trend for lower values at higher levels of plaque coverage could only be found for Primescan and Trios 5, but the trend was less than without correction. The limits of agreement were markedly narrower than without correction, ranging from -12.6 to -13.8 for the lower limit and 13.3 to 14.3 for the upper limit (Fig. 6, Table 3).

At the participant level and for both the uncorrected (Table 4) and corrected (Fig. 7, Table 4) values, the Bland-Altman analysis showed neither a significant systematic nor significant proportional bias for the test IOS compared to the control IOS. However, Primescan showed a visual tendency towards lower values at higher plaque coverage levels, but this did not reach significance ($p = 0.957$). The limits of agreement for uncorrected values were between -11.2 and -9.1 for the lower limit and between 11.4 and 12.6 for the upper limit.

After correction, a clearly improved agreement between the test and control IOS was achieved, similar to the values of the individual teeth. The limits of agreement ranged between -6.1 and -5.4 for the lower limit and between 5.3 and 7.0 for the upper limit.

3.2. Reduction of P% values after intervention

The P% and P%_{corr} values for T1 and T2 are listed in Table 2. The control IOS and all three test IOS detected a significant reduction in plaque coverage after habitual toothbrushing, both with the uncorrected and corrected values and both at the tooth level ($n = 120$) and at the

Table 2
Uncorrected (P%) and corrected (P%_{corr}) values (mean $\pm$ SD) for plaque coverage at the tooth level after 72 h without oral hygiene (T1) and after habitual toothbrushing (T2). The asterisks indicate significant difference of the respective test IOS to the control IOS (rows), the diamonds indicate significant differences from T1 to T2 for P% and P%_{corr} (columns) respectively; significance level after Bonferroni correction = 0.004.

	3800	i700	Primescan	Trios 5
P% T1	32.6 $\pm$ 22.1	31.2 $\pm$ 24.1	30.1 $\pm$ 20.8	32.8 $\pm$ 22.0
P% T2	18.3 $\pm$ 16.7 ♦	16.3 $\pm$ 15.7 *♦	20.7 $\pm$ 19.0 *♦	17.0 $\pm$ 15.7 ♦
P% _{corr} T1	31.8 $\pm$ 21.0	31.5 $\pm$ 22.4	31.1 $\pm$ 20.8	32.3 $\pm$ 21.6
P% _{corr} T2	18.2 $\pm$ 16.4 ♦	16.8 $\pm$ 15.9 *♦	19.2 $\pm$ 17.5 ♦	17.3 $\pm$ 15.3 ♦

participant level ($n = 10$) when the overall value for P% of all values was considered. The p-value was < 0.001 for all cases.

Similar to the control IOS, the test IOS showed different levels of plaque coverage and effects of brushing depending on the tooth type and surface area (Table 5). On the oral surfaces, there were no substantial differences between tooth types after 72 h without oral hygiene, with median P%_{corr} values slightly greater than 20 %. In the case of molars and premolars, the plaque values were numerically reduced by tooth-brushing, but this was not significant. Plaque reduction in the anterior teeth after Bonferroni correction reached significance only for the i700 and Primescan. The vestibular surfaces showed a different pattern: plaque coverage at T1 was the highest in the molars and decreased from the premolars to the incisors. Brushing significantly reduced plaque coverage in all tooth types.

4. Discussion

The present methodological study aimed to answer the question of whether IOS types, aside from the control IOS that we previously used in earlier studies [11,16], are suitable for the detection and quantification of disclosed plaque. Ideally, the use of IOS as a plaque detection tool should cover a wide range of applications; thus, the test IOS should not only reliably identify plaque at a certain point in time but also changes in plaque amount due to interventions such as oral hygiene measures. Therefore, we asked the participants to abstain from oral hygiene for 72 h. The absence of oral hygiene measures for 72 h had no negative effects on oral health. Signs of gingivitis only appear after more than five days without oral hygiene [21] and are reversible after plaque removal [22]. As oral hygiene measures were resumed at the end of the second session, no disadvantages were expected for the participants. However, after this period of undisturbed plaque growth, we expected relatively high levels of plaque coverage, from which we could measure the reduction in plaque after toothbrushing. This also allowed for the identification of potential proportional and systematic biases across a wide range of plaque levels. In addition, we not only looked at an aggregate measure for each participant, which might be used in oral hygiene studies analogous to a score from a plaque index, but also performed an analysis at the tooth level. The consideration of individual tooth surfaces could be relevant for plaque distribution and growth analyses as well as for investigations that pursue a detailed analysis of the effectiveness of oral hygiene techniques, for example, in different regions of the dentition.

Control IOS provided very evenly illuminated images of tooth surfaces with good colour differentiation from stained plaque, allowing batch processing for convenient and fast evaluation of a large number of images with minimal manual correction. Because batch processing is an important component of planimetric evaluation, we also considered this feature when evaluating the test IOS.

The test IOS provided different visual colour representations with yellowish or bluish tones. This probably resulted from the different acquisition technologies or filters used by the different IOS scanning software. After the first analysis, it became clear that the threshold value for the delimitation of pixels assigned to the clean and plaque-covered tooth surfaces used for the control IOS was not suitable for the test IOS. When comparing the planimetrically determined plaque-covered areas with those identified visually, this threshold led to a systematic overestimation or underestimation of plaque coverage. However, a suitable threshold value was determined for each individual test IOS. Analysis software should be designed to have the option to easily adjust to thresholds. Once the thresholds were set for each IOS, batch processing was possible. Nevertheless, in some individual images, the planimetrically determined plaque area did not correspond to the visual impression. In these cases, planimetric analysis was repeated with a manually adjusted threshold value until agreement with the visual impression was achieved. This manual correction was particularly extensive for Primescan.

The analyses were performed in the RGB colour space to enable clear

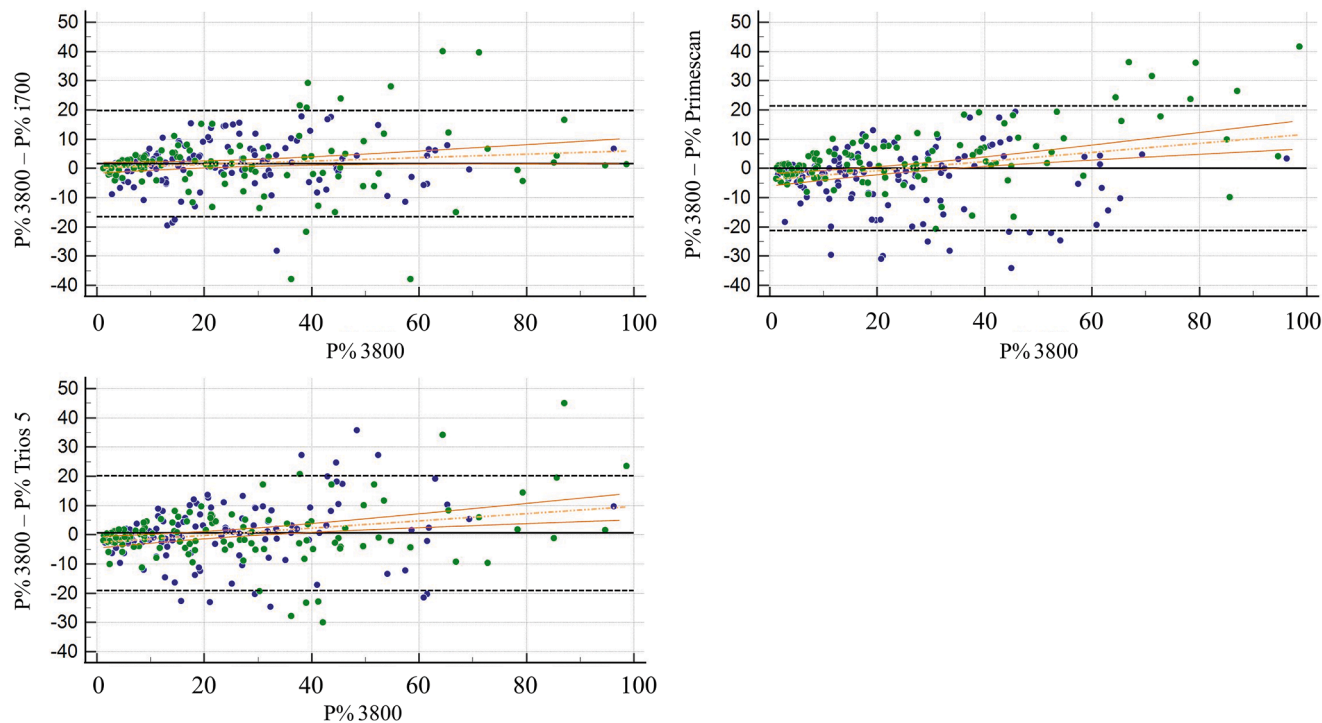


Fig. 5. Bland-Altman plots showing the difference between the P% values of the control IOS and the test IOS at the tooth level ($n = 240$). Solid black line: mean difference, dashed black line: 95 % CI. Double dashed orange line = regression line with 96 % CI (solid orange line). The regression analysis revealed a significant trend towards lower values of all test IOS compared to the control IOS (i700: $p = 0.04$; Primescan: $p < 0.0001$; Trios 5: $p = 0.0001$). Blue circles: vestibular surfaces, green circles: oral surfaces.

Table 3

Bland-Altman analysis at the tooth level ($n = 240$) for systematic bias (mean difference from the P% values of the control IOS with one test IOS each) and the lower and upper limits of agreement with 95 % CI. The analysis was performed for the uncorrected and corrected planimetric values.

	Mean Difference	Lower limit	Upper limit
uncorrected			
i700	1.7 (0.5; 3.0) $p = 0.004$	-16.4 (-18.4; -14.4)	19.9 (17.8; 21.9)
Primescan	0.1 (-1.3; 1.5) $p = 0.921$	-21.3 (-23.6; -18.9)	21.4 (19.0; 23.8)
Trios 5	0.6 (-0.7; 1.8) $p = 0.374$	-19.0 (-21.2; -16.8)	20.1 (18.0; 22.3)
corrected			
i700	0.8 (-0.04; 1.7) $p = 0.061$	-12.6 (-14.1; -11.1)	14.3 (12.8; 15.8)
Primescan	-0.16 (-1.0; 0.7) $p = 0.725$	-13.6 (15.1; -12.1)	13.3 (11.8; 14.8)
Trios 5	0.2 (-0.7; 1.1) $p = 0.339$	-13.8 (-15.4; -12.3)	14.2 (12.7; 15.8)

discrimination of plaque-covered areas for the control IOS. This method appears to be particularly suitable if the images have a wide colour spectrum. However, it is also possible to use other colour spaces such as the hue, saturation, and intensity (HSI) colour space [8]. The Primescan images in particular were characterised by a very narrow RGB colour spectrum with a dominance of red and green in both plaque-covered and plaque-free areas, with varying colour values between different images, which explains the variable threshold values in the images from this IOS. In such cases, the HSI colour space was likely to be more appropriate. A software-generated conversion of RGB to HSI or to the device-independent lab colour space could offer extended evaluation options [23]; however, more research is necessary on this topic.

Although the image quality of the test IOS visually appeared quite

different, the plaque coverage values agreed well with the control IOS; thus, the null hypothesis could be at least partially accepted. Even without corrections, the plaque levels were of the same order of magnitude with all three test IOS, as well as with the control IOS. The limits of agreement were relatively wide; however, there was a proportional bias towards smaller measurements of the test IOS at higher P % values, especially with Primescan in the vestibular areas. After correction, the agreement margins were clearly narrowed and the proportional bias was reduced; the latter only reached significance for Primescan and Trios 5. The picture was similar at the participant level, where the limits of agreement were narrower for the uncorrected values than at the tooth level.

The degree of plaque coverage, as well as the effects of toothbrushing, vary in different areas of the dentition, and IOS plaque quantification should map this. Regarding the vestibular surfaces, there were relatively large differences between the tooth groups, with the molars showing the greatest plaque coverage and the anterior teeth showing the lowest. In contrast, plaque distribution on the oral surfaces seems more homogeneous [10,24]. When oral hygiene is suspended after professional teeth cleaning, the greatest plaque regrowth occurs on the vestibular surfaces of the maxillary molars and decreases towards the anterior teeth; this tendency is not as obvious on the oral surfaces [10]. Although the vestibular surfaces partly achieved higher plaque coverage than the oral surfaces after a period without oral hygiene, toothbrushing resulted in greater plaque reduction and overall lower plaque coverage rates on the vestibular surfaces than on the oral surfaces. This may be explained by the finding that individuals generally brush the vestibular surfaces much longer than the oral surfaces, which are often only reached incompletely or not at all [25,26]. The plaque distribution pattern described above, after suspending oral hygiene and subsequent toothbrushing, was also confirmed in the present study. This was evident not only from the data of the control IOS but from those of

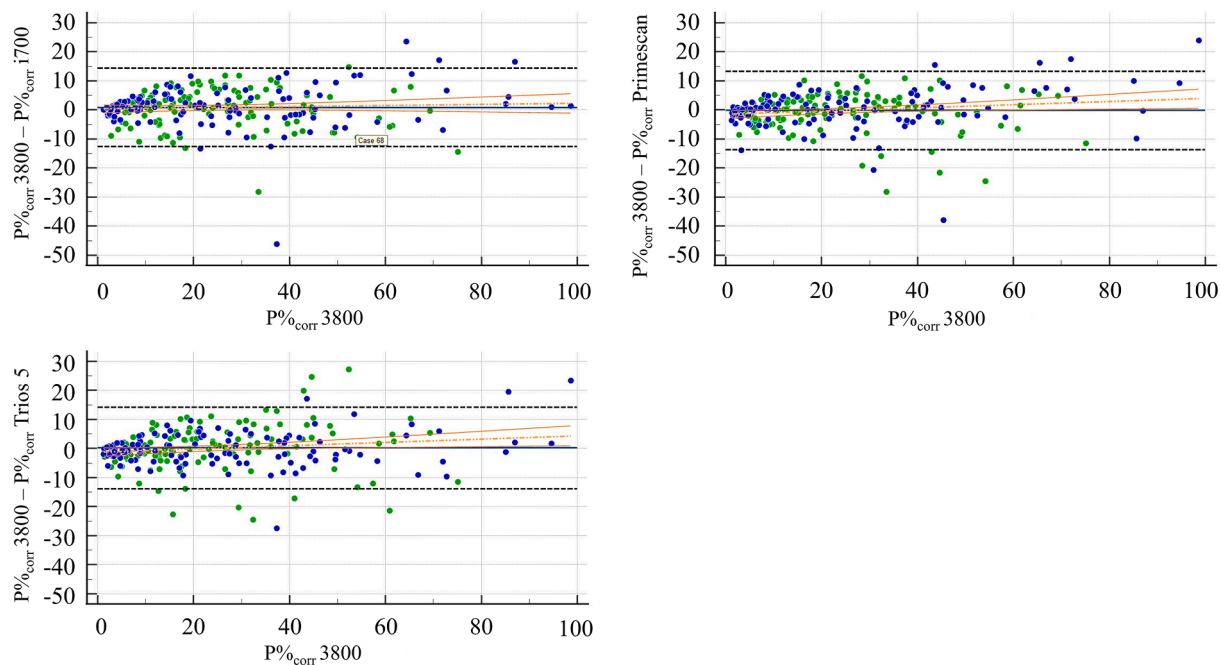


Fig. 6. Bland-Altman plots showing the difference between the corrected P% values ($P\%_{\text{corr}}$) of the control IOS and the test IOS at the tooth level ($n = 240$). Solid black line: mean difference, dashed black line: 95 % CI. Double dashed orange line = regression line with 96 % CI (solid orange line). After correction, the significant trend towards lower values of the test IOS compared to the control IOS remained only for Primescan and Trios 5 (i700: $p = 0.38$; Primescan: $p = 0.01$; Trios 5: $p = 0.01$). Blue circles: vestibular surfaces, green circles: oral surfaces.

Table 4
Bland-Altman analysis at the participant level ($n = 20$) for systematic bias (mean difference from the P% values of the control IOS with one test IOS each) and the lower and upper limits of agreement with 95 % CI. The analysis was performed for the uncorrected and corrected planimetric values.

	Mean Difference	Lower limit	Upper limit
uncorrected			
i700	1.7 (−0.9; 4.3) $p = 0.182$	−9.1 (−13.6; −4.6)	12.6 (8.1; 17.1)
Primescan	0.07 (−2.6; 2.8) $p = 0.957$	−11.2 (−15.9; −6.5)	11.4 (6.7; 16.1)
Trios 5	0.6 (−2.1; 3.2) $p = 0.652$	−10.4 (−15.0; −5.9)	11.6 (7.0; 16.1)
corrected			
i700	0.8 (−0.6; 2.3) $p = 0.252$	−5.4 (−7.9; −2.8)	7.0 (4.5; 9.6)
Primescan	−0.2 (−1.5; 1.1) $p = 0.403$	−5.6 (−7.9; −3.4)	5.3 (3.0; 7.6)
Trios 5	0.2 (−1.3; 1.7) $p = 0.791$	−6.1 (−8.7; −3.5)	6.5 (3.9; 9.1)

all test IOS.

The visualisation of disclosed plaque with IOS is interesting for research purposes and can be helpful in the context of oral hygiene education. For this, however, it is important that the dentition is depicted as photo-realistically as possible, and that plaque areas can be distinguished from the clean tooth surface as sharply and as precisely as possible. Currently, this seems to be less possible with the test IOS than with the control IOS. Because IOS technology and the field of application of these devices are developing rapidly, better image quality should be provided with all IOS models in the future.

The fact that only younger individuals with well-formed dental arches and tooth-coloured restorations, without pronounced gingival recession or wide interdental areas, were included can be regarded as a limitation of this study. However, these conditions provide optimal scanning conditions without artefacts or scan holes and therefore supply optimal parameters for plaque quantification. Whether similar results can be achieved when patients have metal restorations, extended gingival recessions, and wide interdental areas and how different IOS cope with such conditions remain to be investigated.

The ability to digitally record plaque using various IOS and to quantify the amount of plaque planimetrically, as presented in this study, offers many advantages.

First, it may minimise clinical time and allow remote data collection by clinical research staff who are familiar with and trained in IOS but are not necessarily qualified dentists. This allows for multicentre studies without the need for time-consuming index calibration. Additionally, a clear separation between data collection and analysis was achieved, which effectively eliminated potential bias. Furthermore, the digital storage of data allows any information to be reproduced at a later date or, in accordance with the ethical vote and data protection requirements, re-analysed at any time from different perspectives and questions using different methods. Finally, it facilitates the verification of data processing and ensures the integrity and accuracy of the data, which are crucial for valid conclusions and informed clinical decisions.

5. Conclusions

There was good agreement between the plaque coverage values measured using the three test IOS (i700, Primescan, and Trios 5) and the control IOS (3800) at all levels of analysis. Automated differentiation of

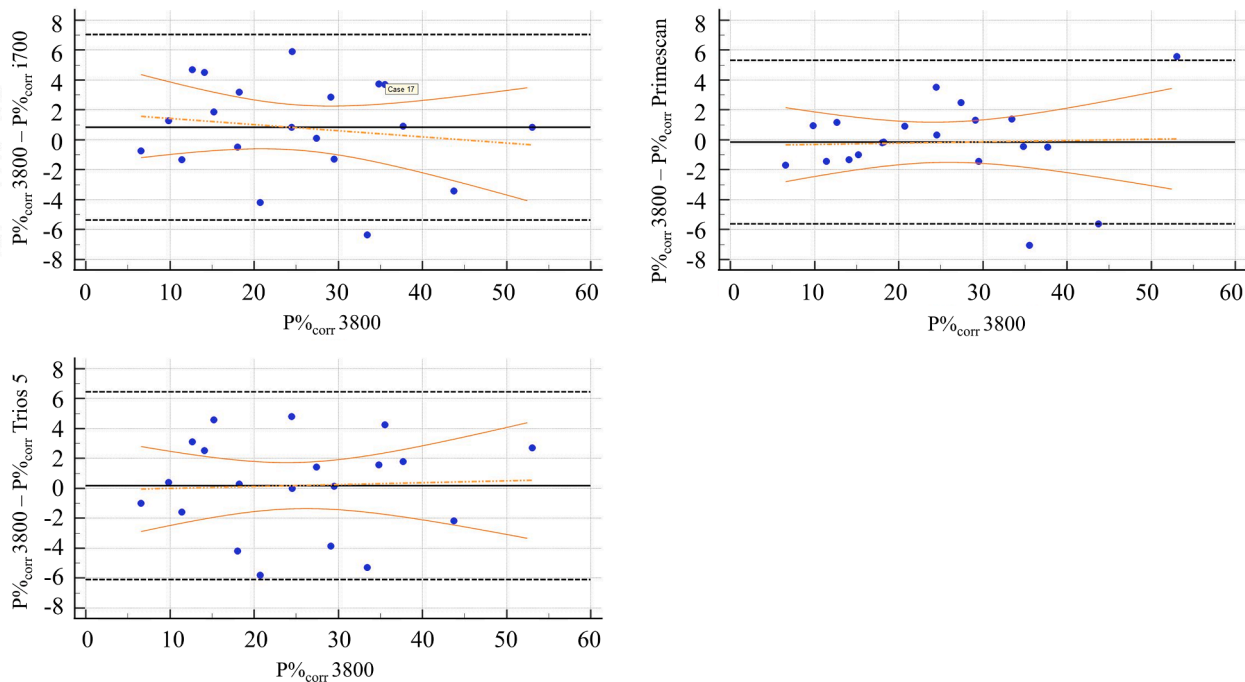


Fig. 7. Bland-Altman plots showing the difference between the corrected P% values (P%corr) of the control IOS and the test IOS at the participant level ($n = 20$). Solid black line: mean difference, dashed black line: 95 % CI. Double dashed orange line = regression line with 96 % CI (solid orange line). The regression analysis revealed no significant trend as a function of the P% values (i700: $p = 0.50$; Primescan: $p = 0.87$; Trios 5: $p = 0.84$).

Table 5

P%_{corr} (median [95 % CI]) after 72 h without oral hygiene (T1) and after habitual toothbrushing (T2) by surface area and tooth type; p-values refer to the difference between T1 and T2. Significance level after Bonferroni correction < 0.002.

		3800	i700	Primescan	Trios 5
oral					
Molars	T1	24.3 (13.5; 30.5)	20.0 (11.7; 24.4)	21.7 (12.5; 29.1)	23.9 (13.9; 27.5)
	T2	19.5 (12.1; 30.2)	14.3 (11.1; 28.8)	16.2 (13.0; 30.3)	15.5 (11.0; 24.5)
		$p = 0.263$	$p = 0.218$	$p = 0.825$	$p = 0.391$
Premolars	T1	24.1 (13.0; 35.2)	23.0 (10.9; 32.2)	19.7 (13.8; 33.9)	23.3 (18.4; 32.9)
	T2	15.7 (12.8; 30.6)	17.1 (12.6; 30.6)	19.4 (12.8; 25.7)	17.1 (12.9; 21.6)
		$p = 0.040$	$p = 0.107$	$p = 0.107$	$p = 0.008$
Incisors	T1	20.0 (17.2; 33.0)	26.8 (15.0; 36.0)	22.4 (17.0; 34.9)	25.1 (13.5; 32.1)
	T2	11.7 (7.4; 23.6)	14.1 (9.5; 18.7)	14.4 (11.1; 24.4)	14.7 (9.4; 21.8)
		$p = 0.004$	$p < 0.001$	$p < 0.001$	$p = 0.008$
vestibular					
Molars	T1	45.1 (39.9; 54.1)	41.1 (35.4; 47.6)	45.0 (37.3; 52.8)	47.7 (37.3; 55.1)
	T2	14.7 (8.8; 26.2)	8.9 (6.1; 22.2)	12.3 (7.1; 27.3)	11.8 (8.1; 26.5)
		$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
Premolars	T1	29.3 (21.8; 41.2)	33.0 (22.0; 42.3)	28.7 (22.0; 40.8)	32.3 (20.8; 42.4)
	T2	8.3 (4.4; 17.1)	8.0 (4.4; 18.4)	8.3 (6.2; 16.9)	8.9 (6.2; 16.9)
		$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$
Incisors	T1	18.3 (9.4; 36.7)	17.7 (6.8; 46.9)	15.9 (7.0; 37.7)	18.3 (12.3; 41.0)
	T2	3.8 (2.5; 8.8)	3.9 (2.0; 5.6)	4.8 (2.9; 12.7)	4.4 (2.4; 6.5)
		$p < 0.001$	$p < 0.001$	$p < 0.001$	$p < 0.001$

pixels belonging to the plaque-covered tooth surface from those belonging to the clean tooth surface was less reliable for the test IOS than for the control IOS when analysed in the RGB colour space, requiring visual verification and manual adjustment. An image analysis method more specific to the type of IOS may optimise the results.

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CRediT authorship contribution statement

Katja Jung: Writing – original draft, Visualization, Validation, Investigation, Conceptualization. **Katja Giese-Kraft:** Writing – review & editing, Conceptualization. **Maximiliane Amelie Schlenz:** Writing – review & editing, Investigation, Conceptualization. **Bernd Wöstmann:** Writing – review & editing, Resources. **Carolina Ganss:** Supervision, Project administration, Investigation, Conceptualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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